

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CE/IT) Examination

November 2013

MA 201 Discrete Mathematics

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Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Consider standard terminology, attempt following: [07]

1. Give an example of a partial order set with no maximal element.
2. Define equivalence relation.
3. Give an example of distributive lattice.
4. Define fundamental product.
5. Statement of Stone's Representation theorem.
6. A Boolean expression containing sum of minterms is known as _____.

a. Sum of product canonical form	b. Product of sum canonical form	c. Symmetric expression	d. none
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7. How many input combinations are required in a truth table, when number of input variables is 'n'?

a. $2n$	b. 2^n	c. n^2	d. none of these
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Q - 2 (a) Is this true that every finite lattice has Hasse diagram representation? Justify. Draw [05]
Hasse diagram for $X = \{2, 4, 5, 6, 8, 10, 12, 24\}$ and define a relation ' $\leq$ ' on X by $x \leq y \Leftrightarrow 2x|y$.

(b) Define POSET. Define Identity relation on $A = \{1, 2, 3\}$. Is this identity relation partial [05]
order relation? Is it equivalence relation? Justify.

(c) Define minimal elements, least element. Let $A = \{2, 3, 4, 6, 8, 12, 24, 48\}$ and ' $\leq$ ' be the partial [04]
order of divisibility. Find our minimal, maximal, least and greatest elements of A

OR

Q - 2 (a) Consider universe of discourse as all people in the world. For x and y , define [05]
 $L(x, y) =$ "person x loves y ", translate following English sentences in to logical expressions: (1) Everybody loves 'z'. (2) Everybody loves somebody. (3) There is somebody whom everybody loves. (4) Nobody loves everybody (5) There is somebody whom nobody loves

(b) Find PCNF and PDNF of an expression $(\sim P \rightarrow R) \wedge (Q \leftrightarrow P)$ using truth table. [05]

(c) Define Sub-Boolean algebra. Find out all possible Sub-Boolean algebra of $(S_{30}, *, \oplus)$ [04]

Q - 3 (a) Define Boolean Algebra. In any Boolean algebra prove that $a = b \Leftrightarrow a'b + ab' = 0$ [04]

(b) What is k-map? Draw circuit for expression $f(a, b, c) = \Pi(0, 2, 4, 5)$ using k-map [05]

(c) Construct argument to show that the following premises imply the conclusion "it rained." [05]

"If it does not rain or if there is no traffic dislocation, then the sports day will be held and the cultural programme will go on", "if the sports day is held, the trophy will be awarded" and "the trophy was not awarded".

OR

Q - 3 (a) Define complemented lattice. Show that every element in a Boolean algebra has unique complement. [05]

(b) State and prove D'morgan's properties for lattices. [05]

(c) Determine validity of following arguments: [04]

(i) If you study well then you will pass in your examination. Not study well;
You will not pass in your examination.

(ii) $(p \rightarrow r) \vee (q \rightarrow r) \vee (\neg(p \vee q)) \vdash \therefore \neg r$

SECTION - II

Q - 4 Consider standard terminology, attempt following: [07]

1. Draw a complete graph with five vertices.
2. State first theorem of graph theory.
3. True or false, " $K_{3,3}$ is non-planar graph"
4. True or False, "A tree with 10 vertices has precisely 9 numbers of edges."
5. True or false, "If path does not have any repetition of vertices, path is known as elementary path."
6. True or false, "All the entries in a path matrix are either zero or one."
7. A complete graph with four vertices has _____ number of spanning trees.

a. 16	b. 10	c. 12	d. 06
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Q - 5 (a) Use FIGURE 3, to answer following questions: [08]

- a. Find an elementary path of length 4.
- b. Is graph regular?
- c. Is graph Mixed?
- d. Find a cycle of length 7, if exists
- e. Find out degrees of all the vertices.
- f. List out all pendants for given graph.
- g. Is graph connected?

(b) Define planar graph. Write statement of Kuratowski's theorem and then prove that K_5 is non-planar. [06]

OR

Q - 5 (a) Using matrix representation only answer the following questions:

[08]

- Find degree of every vertex, total number of edges and total number of loops.
- Find out total number of paths of length 3.
- Verify that graph is connected or not.
- Find out corresponding path matrix.

$$A(G) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

(b) Define: Complete graph. Complete Bipartite Graph, Isomorphic graphs

[06]

Give an example of complete graph which is isomorphic to complete bipartite graph

Q - 6 (a) Define minimal spanning tree. Find out minimal spanning tree for the network model given in **FIGURE 1** using Krushkal's Algorithm. [07]

(b) Use **FIGURE 2**, and find shortest route from origin O to destination T, using Dijkstra's algorithm. [07]

OR

Q - 6 (a) Draw graphical tree expression for tree given below. Find out all descendants of 'b'. Find out sibling of 'd'. Find height of a tree. Is it binary tree? [07]

$$(a(b(d)(e(f)(g)))(c(h(i)(j)(k)))(l(m(n))))$$

(b) Define cut vertex, vertex connectivity, Euler Graph with illustration. Draw a graph with 4 vertices and 8 edges, which is Hamiltonian but not planar. [07]

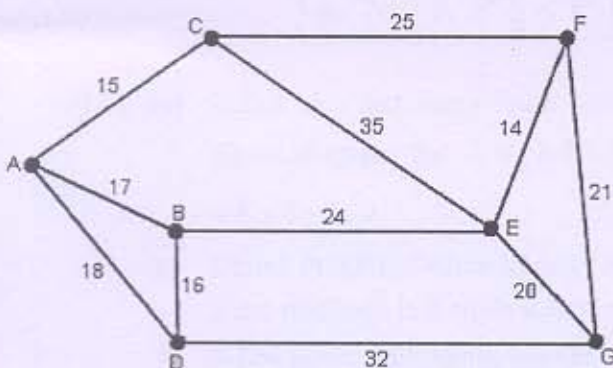


Figure 1

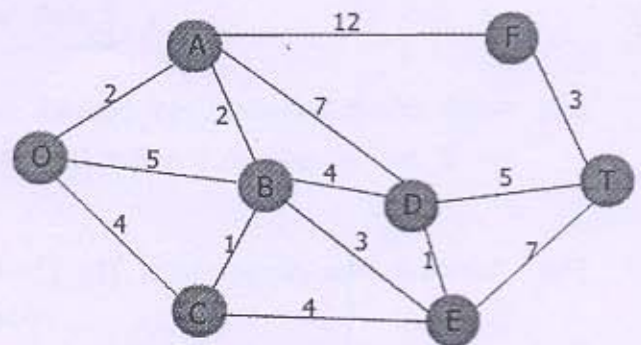


Figure 2

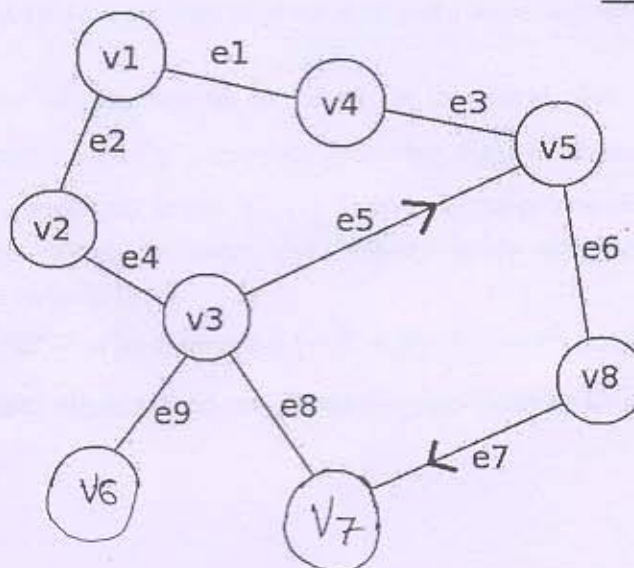


Figure - 3